

Dietary Approaches to Kidney Diseases

Nimrit Goraya | Donald E. Wesson

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INTRODUCTORY POINTS

DIET IS A KEY COMPONENT OF MANAGEMENT OF CHRONIC KIDNEY DISEASE

Analysis of U.S. data showing that diet is the single largest chronic kidney disease (CKD)-related risk factor for death and disability in individuals with CKD highlights the importance of diet in CKD management.¹ Furthermore, a “healthy” diet in patients with CKD is associated with reduced mortality.²⁻⁴ These data support that diet can enhance the length and quality of life of individuals with CKD and should encourage clinicians to know and recommend food patterns that accomplish these goals. Additional reasons for clinicians to consider diet as the cornerstone of CKD management include the following:

- Diet is a cornerstone component of the management of diabetes, the major cause of CKD in developed societies.⁵ Individuals with diabetes-related CKD must have their diabetes properly managed along with their CKD. Importantly, the “healthier” (we will subsequently detail the components of a “healthy” diet) the diet in CKD due to type 2 diabetes (T2D), the lower the mortality and the lower the risk for CKD progression.⁶
- Diet is recommended as first-line management of hypertension,⁷ a comorbid condition present in more than 90% of individuals with CKD.⁵ Consequently, diet will help control hypertension in CKD and long-term studies support that better hypertension control yields lower mortality in CKD.^{8,9}

- Diet helps to control some of the metabolic complications of CKD.
- Diet also helps ameliorate some of the organ dysfunction associated with CKD.
- Properly applied, diet can limit body accumulation of waste products ordinarily excreted by the kidney that cause harm, thereby delaying the need for kidney replacement therapy in the form of dialysis or kidney transplantation. The accompanying imbalance in acid-base, fluid, electrolytes, and mineral bone disease decreases the quality of life of patients with CKD. Ameliorating these metabolic complications by skillful application of effective food patterns that successfully delay the need for kidney replacement therapy can improve patient life quality and reduce individual patient and societal costs for CKD management.
- Diet might also be used to help slow CKD progression as will be discussed.

Together, these data support that diet in CKD plays an important role in its prevention, its progression once it is established, management of its complications, and lowering its associated mortality. Although the focus of our discussion will be strategies and tactics available to clinicians caring for patients with CKD, the data will support that changes in food policy will facilitate some interventions, particularly when considering strategies for CKD prevention. **The data to be discussed support diet as the foundational, “food first”¹⁰ approach to the management of individuals with CKD, as well as those at risk for it, with pharmacologic therapy as adjunctive to diet. This is the reverse of the**

current approach in which pharmacologic therapy is foundational to CKD management. We begin our discussion with the contributions of diet to the major disease entities responsible for CKD, diabetes, and hypertension, which account for almost two-thirds of incident end-stage kidney disease (ESKD) cases in the United States.⁵

DIETARY CONTRIBUTIONS TO MAIN SYNDROMES THAT CAUSE CHRONIC KIDNEY DISEASE: DIABETES AND HYPERTENSION

DIABETES

Diabetes mellitus remains the single largest contributor to ESKD in the United States with type 2 diabetes being responsible for more than 90% of the cases and type 1 diabetes (T1D) contributing the remainder.^{5,11} Although U.S. incidence of T2D is decreasing, its prevalence is increasing,¹² likely due to individuals with T2D living longer, which unfortunately exposes them to higher risk of developing CKD. Ten years after T2D diagnosis, 5.3% had macroalbuminuria (i.e., >300 mg of daily urine albumin excretion⁵), indicative of established nephropathy.¹³ Unfortunately, many patients with T2D already have signs of CKD at the time of diagnosis,^{14,15} possibly because they had experienced an indeterminate period of the abnormal metabolic state known as prediabetes, which can be associated with signs of kidney injury, like albuminuria.¹⁶ More concerning, about one-third of U.S. adults have prediabetes¹² and such individuals have a higher prevalence of reduced estimated glomerular filtration rate (eGFR) compared with the general population.¹⁶ It is therefore important to consider dietary approaches to CKD, particularly strategies intended to prevent CKD onset and its progression, within the context of dietary contributions to this major health condition that leads to CKD in developed societies.

DIETARY CONTRIBUTIONS TO TYPE 2 DIABETES MELLITUS

Overweight is the single most important modifiable risk factor for T2DM,¹⁷ so dietary factors that contribute to it thereby provide the context for diabetes-related CKD. Observational studies support that adherence to a “healthy” as opposed to an “unhealthy” diet reduces the risk for T2D. Such preventive diets included the Dietary Approaches to Stop Hypertension (DASH) diet and the Mediterranean diet, which are each associated with reduced T2D risk.^{18,19} The DASH diet emphasizes fruits and vegetables, whole grains, fat-free or low-fat dairy products, fish and poultry rather than red meat, beans, nuts, and vegetable oils. The Mediterranean diet emphasizes fresh fruits and vegetables, olive oil, and fish as the source of animal protein and deemphasizes simple sugars and red meat as a protein source.

Dietary components that appear to influence T2D risk include:

- *Character of ingested carbohydrates.* High intake of the indigestible component of complex carbohydrates known

as fiber is associated with reduced T2D risk.^{20,21} Higher fiber intake enhances the growth of gut bacteria whose fermentation of the ingested fiber produces short-chain fatty acids that promote improved glycemic control.²² Whole-grain foods are particularly high in fiber, and fruits and vegetables in general are fiber sources. On the other hand, chronic and excess ingestion of foods with a greater propensity to increase blood glucose concentrations when ingested (i.e., foods with a high glycemic index like simple sugars) is associated with increased T2D risk.²³⁻²⁵

- *Character of ingested fat.* Although high intake of saturated fat appears not to affect the risk for T2D,²⁶ high intake of the polyunsaturated fats omega 6^{27,28} and omega 3²⁹ appear to reduce the risk for T2D.
- *Character of ingested protein.* High intake of animal-based protein is associated with higher risk for T2D,^{30,31} particularly increased intake of red meat.³²⁻³⁵ On the other hand, diets proportionately high in plant-based protein are associated with lower T2D risk.³⁶
- *Dietary effects on endogenous acid production.* Animal-based protein is metabolized to yield acid, but most plant-based protein is metabolized to yield base or is metabolized to yield neither.³⁷ In addition, a high-salt (NaCl) diet increases endogenous acid production.³⁸ Whether ingested diets are acid producing or base producing is determined to a large degree by the balance of acid-producing and base-producing dietary components. Fig. 55.1 compares the acid- and base-producing capacities of common foods. High acidogenic diets increase insulin resistance,³⁹ their chronic ingestion is associated with increased T2D risk,⁴⁰⁻⁴² and insulin resistance in T2D is associated with hypertension and microalbuminuria.⁴⁰
- *Dietary effects of ultraprocessed foods (UPFs).* Ultraprocessed foods have poorer nutritional quality including higher content of saturated fat, added sugar, energy density, sodium chloride (NaCl or salt), and increased levels of contaminants from processing or packaging and food additives, all of which negatively impact health. Studies show a positive association between UPF consumption and new-onset CKD in participants with diabetes compared with those without diabetes.⁴³ Data from the Atherosclerosis Risk in Communities cohort showed a linear relation between UPF intake and risk of developing CKD; furthermore, the data showed that substituting one serving per day of UPFs with minimally processed foods was associated with a 6% lower risk of CKD.⁴⁴ Unfortunately, intake of UPF is increasing in the United States.⁴⁵

Together, these data guide dietary strategies to prevent diabetes, particularly in individuals and populations at increased risk. In addition, they help guide dietary management of patients with CKD and concomitant diabetes.

HYPERTENSION

Hypertension is the second largest contributor to ESKD in the United States and is a comorbid factor in more than 90% of patients with CKD.⁵ As discussed for diabetes, it is important to consider dietary approaches to CKD, particularly strategies intended to prevent CKD, its

the DASH diet and is already higher in vegetables than diets typical of developed societies, was associated with an even lower risk for development of hypertension.⁶⁷ Furthermore, reduction of red meat consumption and substitution with vegetable protein reduced blood pressure in prehypertensive, postmenopausal women.⁶⁸

- *Dietary effects of endogenous acid production on hypertension.* As indicated, the body metabolizes animal-based protein in a way that increases endogenous acid production, whereas most plant-based protein is metabolized in a way that either decreases endogenous acid production or has no net effect on it.³⁷ In addition, as indicated, high-salt diets increase endogenous acid production.³⁸ Studies in children⁶⁹ and adults^{70,71} support that diets which increase endogenous acid production increase blood pressure. Because dietary sodium increases,³⁸ dietary potassium decreases,³⁷ animal-based protein increases,³⁷ and plant-based protein decreases³⁷ endogenous acid production, dietary effect on endogenous acid production might be a common mechanism through which these and possibly other dietary components influence blood pressure and the risk for hypertension.
- *Obesity.* There are, of course, strong dietary contributions to obesity and population studies show that obesity is associated with increased risk for hypertension.^{72,73} In addition, weight loss in obese persons with hypertension generally reduces their blood pressure.⁷⁴

Strategies to reduce CKD incidence in populations can begin by reducing the incidence of its two major causes, diabetes (particularly T2D) and hypertension. Data support strong dietary contributions to both and suggest that dietary strategies instituted for individual at-risk patients and, more broadly, populations as a whole, hold promise to reduce CKD incidence through reducing the incidence of diabetes and hypertension. Such interventions might be more effectively instituted at an overall population rather than on an individual level, particularly when focused on population segments at particularly high CKD risk. Furthermore, because diabetes is the single largest contributor to CKD in developed societies and because more than 90% of patients with CKD have hypertension, these data support effective dietary strategies in individual CKD patients to manage concomitant diabetes and/or hypertension in patients with CKD.

DIETARY CONTRIBUTIONS TO CHRONIC KIDNEY DISEASE INCIDENCE AND PROGRESSION

DIETARY FACTORS ASSOCIATED WITH INITIATION OF CHRONIC KIDNEY DISEASE

Similar to what has been described for the two syndromes that are the main contributors to CKD, observational studies support that adherence to a “healthy” as opposed to an “unhealthy” diet is associated with reduced risk for CKD.⁷⁵⁻⁷⁸ Although the components of a healthy diet that mediate reduced CKD risk remain to be clearly identified, observations that adherence to the Mediterranean diet which emphasizes fresh fruits and vegetables, olive oil, and fish as the source of animal protein and deemphasizes red

meat and simple carbohydrates is associated with reduced CKD risk⁷⁹⁻⁸² offers some dietary elements to consider.

Plant-based diets are high in fiber and such diets, particularly those with a high ratio of fiber-to-animal-based protein, are associated with lower CKD risk,^{82,83} possibly in part because diets with a high proportion of animal-based protein are associated with high levels of gut-derived substances that are putative kidney toxins.⁸⁴ As indicated, most plant-based protein is metabolized to yield base after ingestion as opposed to animal-based protein, which yields acid.³⁷ Lower-acid diets are associated with lower CKD risk, and higher acid diets are associated with higher CKD risk.^{85,86} Higher intake of red meat in particular is associated with higher CKD risk.^{87,88} Higher acid-producing diets, caused largely by disproportionately higher intake of animal-based protein, might contribute to kidney injury that initiates CKD by causing chronic hyperfiltration,⁸⁹ a phenomenon linked to hemodynamically induced kidney injury.⁹⁰

Other potential kidney-protective benefits of the Mediterranean diet might derive from its oily fish and other plant-based foods that contain omega-3 polyunsaturated fatty acids, high intake of which has been associated with reduced CKD risk.⁹¹ Omega-3 polyunsaturated fatty acids are also associated with reduced risk of obesity,⁹² and obesity itself is associated with increased CKD risk,^{93,94} even independent of diabetes and hypertension.^{95,96}

DIETARY FACTORS ASSOCIATED WITH PROGRESSION OF ESTABLISHED CHRONIC KIDNEY DISEASE

Individuals with albuminuria, even with initially normal eGFR, experience higher risk for subsequent eGFR decline⁵ whether due to diabetes⁹⁷ or nondiabetic CKD associated with primary hypertension.⁹⁸ As has been described for diabetes, hypertension, and initiation of CKD, adherence to a “healthy” as opposed to an “unhealthy” diet is associated with slower eGFR decline of established CKD.^{6,99,100} These “healthy” diets have high content of plant-based protein,⁷⁵⁻⁷⁸ and increasing the proportion of dietary plant-based protein decreases urine indices of kidney injury¹⁰¹ and slows CKD progression.¹⁰²⁻¹⁰⁵ Because, as indicated, plant-based protein is largely metabolized to yield base, at least part of the benefit of plant-based diets might be their effect to reduce dietary acid^{37,101} because correction of metabolic acidosis¹⁰³⁻¹⁰⁸ and dietary acid reduction even without underlying metabolic acidosis^{109,110} slows the rate of eGFR decline in CKD. Other mechanisms contributed by plant-based protein might be promotion of a gut microbiome that yields reduced metabolism of diet components delivered to the gut into kidney toxic substances.¹¹¹ Diets with higher intake of plant-based components are comparatively lower in sodium than diets higher in animal-based components, particularly processed meats, and higher sodium diets have been associated with more rapid progression of CKD.¹¹²

POTENTIAL MECHANISMS FOR DIET-RELATED KIDNEY INJURY

- **Hyperfiltration.** When a substantial number of functioning nephrons is lost, remaining functioning nephrons increase filtration above that when they were part of a normal contingent of nephrons, a process known as hyperfiltration.⁹⁰ Animal models of CKD support that

the higher glomerular filtration pressure associated with hyperfiltration is injurious to kidneys in the long term and contributes to progressive GFR decline.¹¹³ Higher intake of animal-based protein contributed to hyperfiltration with increased glomerular capillary pressure in these animal models of CKD and diets higher in animal-based proteins induced more severe kidney injury in animals with reduced compared with those with intact nephron mass.^{113,114} Chronically high intake of animal-based dietary proteins also appears to cause hyperfiltration in humans.⁸⁹ Hormonal systems that appear to contribute to hyperfiltration in these animal models of CKD include glucagon, insulin, insulin-like growth factor-1, AII, prostaglandins, and kinins.¹¹⁵⁻¹¹⁸

- **Oxidative stress.** Oxidative metabolism yields molecules that are potentially injurious to surrounding tissue, and innate systems normally remove or detoxify them. These oxygen-related molecules can cause tissue injury if their production exceeds normal detoxifying mechanisms or if the normal detoxifying mechanisms are compromised by a phenomenon known as oxidative stress. The reduced nephron mass animal model of CKD is associated with oxidative stress that is exacerbated by increased intake of animal-based dietary protein.^{119,120}
- **Inflammation.** Progressive nephropathy is characterized by inflammation including fibrosis, particularly of the kidney tubule interstitium.¹²¹ Diets high in animal-based protein increase kidney production of transforming growth factor β -1 and fibronectin, two substances that mediate kidney fibrosis.¹²² Furthermore, high intake of animal-based protein increased kidney levels of endothelin-1, aldosterone, and angiotensin II, each of which mediates progressive GFR decline and tubule-Interstitial fibrosis in the reduced kidney mass model of CKD in animals; on the other hand, increased intake of plant-based dietary protein ameliorated GFR decline and caused less tubule-Interstitial fibrosis.¹²³⁻¹²⁶
- **Acid-induced kidney injury.** As noted, higher dietary acid is associated with increased CKD incidence^{85,86} and progression of established CKD.^{127,128} Animal studies show that high-acid diets increase kidney levels of substances, such as endothelin-1, aldosterone, and angiotensin II, all of which contribute to kidney tubule acidification in response.¹²⁹⁻¹³¹ In turn, this can lead to tubulo-Interstitial injury, further mediating progressive GFR decline.^{123-126,132} Correction of metabolic acidosis in patients with CKD is associated with amelioration of nephropathy progression and reduced urine excretion of these substances.^{103,107,109} Together, these data support that dietary acid contributes to acid-induced kidney injury through increased kidney levels of these, and possibly other, cytokines.

A plant-based diet can ameliorate kidney injury by each of the diet-related mechanisms of kidney injury described.

DIETARY CONTRIBUTIONS TO INCREASED CHRONIC KIDNEY DISEASE-RELATED MORTALITY

Reduced eGFR is associated with increased mortality,¹³³ particularly due to increased cardiovascular mortality,¹³⁴

and was associated with 4% of worldwide deaths in 2013.¹³⁵ The appearance of reduced GFR significantly shortens life expectancy in patients with diabetes compared with those with diabetes and normal GFR.¹³⁶ Diet is the single most important CKD-related risk factor for death and disability,¹ and a “healthy” diet is associated with reduced mortality in CKD.² Excess animal-based protein-to-fiber intake is associated with mortality in patients with CKD,¹³⁷ but increases in dietary fiber were associated with reduced CKD-related inflammation and mortality.¹³⁸ Furthermore, higher fruit and vegetable intake is associated with lower mortality in patients with CKD.¹³⁹

Together, these data support dietary strategies as foundational management of persons at increased risk for or with established CKD in an effort to reduce their risk for developing CKD, for progression of established CKD, and to reduce the excess mortality associated with CKD.

NUTRITIONAL COMPONENTS TO CONSIDER WHEN DESIGNING DIETARY MANAGEMENT OF CHRONIC KIDNEY DISEASE PATIENTS

- **Energy requirements.** Most studies support that energy requirements for patients with CKD are not different from those without CKD, but energy requirements did not decrease in response to decreased caloric intake as they did in non-CKD patients,¹⁴⁰ suggesting that patients with CKD do not respond normally to a reduction in caloric intake. Some studies support a slight increase in energy requirements in patients on hemodialysis due to uremic factors that cause insulin resistance.¹⁴¹ Although patients with CKD might have steady-state levels of energy intake below the recommended 30 to 35 kcal/kg body weight/day,¹⁴² most studies support the 30 to 35 kcal/kg body weight/day recommendation when prescribing a protein-restricted diet for kidney protection to avoid protein-energy wasting.^{143,144} Carbohydrates can account for 5% to 15% of energy requirements for patients with CKD,¹⁴⁵ but practitioners must exercise caution when including foods with high fructose content because it is associated with increased insulin resistance¹⁴⁶ and with increased all-cause and CVD mortality.¹⁴⁷ Additional reasons to limit or avoid dietary fructose as an energy source for patients with CKD is that it increases serum uric acid levels,¹⁴⁸ which are associated with exacerbated nephropathy progression.^{148,149} Furthermore, lowering serum uric acid improves kidney outcomes and reduces the risk for cardiovascular events in patients with CKD.¹⁵⁰
- **Protein requirements.** Empiric observations support minimum dietary protein requirements as 0.6 g/kg of ideal body weight per day; adding 2 standard deviations for a general recommendation to theoretically encompass the dietary protein needs of 97.5% of individuals yields the standard recommendation of 0.75 g/kg bw/day,¹⁵¹ which is often rounded up to 0.8 g/kg/bw/day. This recommendation applies to both vegetarians and nonvegetarians.¹⁵² These guidelines suggest that patients on dialysis increase this amount to 1.2 g/kg bw/day because of their high protein catabolism and/or loss.¹⁵³ Unlike that described for energy requirements, patients with

CKD can adapt to dietary protein restriction by reducing amino acid oxidation and protein degradation.¹⁵⁴ Naturally occurring amino acids contain nitrogen, which, when metabolized, yields nitrogenous wastes that increase blood urea nitrogen (BUN) and possibly adversely affect kidney function.¹⁵⁵

Low-protein diets (0.6–0.8 g/kg bw/day), sometimes supplemented with nonnitrogen keto analogs, have been associated with slower GFR decline^{155,156} and lower levels of gut-derived uremic toxins¹⁵⁷ in patients with nondialysis-requiring CKD. Nevertheless, the largest study to date that examined the effect of total dietary protein restriction on nephropathy progression, the Modification of Diet in Renal Disease study, showed that GFR decline at 3 years was no different among subjects randomized to a low-protein diet (0.58 g/kg/day) and a usual protein diet (1.3 g/kg/day).¹⁵⁸ Furthermore, despite the fact that some studies suggested that low-protein diets could preserve kidney function, some patients with intakes below 0.8 g/kg/bw/day, even the described minimum level of 0.6 g/kg/bw/day, develop protein energy wasting¹⁵⁹ and mortality.¹⁵⁶ Some studies show that patients with CKD can safely eat these diets without compromising nutritional status.^{155,160} Because of the described benefits of primarily plant-based diets, investigators have studied these diets in patients with CKD and show similar nutritional status in such patients compared with those ingesting similar amounts of a primarily animal-sourced protein diet.¹⁶¹

Individuals living in developed societies typically ingest dietary protein in excess of requirements, averaging about 60% more,¹⁵² and most of this ingested protein is animal based.³⁷ As noted, animal-based protein increases intrinsic acid production when metabolized³⁷ and increased dietary acid has been associated with increased risk for CKD appearance^{85,86} and progression.^{127,128} Also as noted, animal-based dietary protein is associated with the production of high levels of gut-derived substances that are putative kidney toxins.⁸⁴ For these reasons, many guidelines recommend restricting dietary protein intake for those living in developed societies to that necessary to satisfy requirements, typically 0.7 to 0.8 g/kg of ideal body weight daily.¹⁴⁵

Ingested dietary protein that is not converted into protein is converted to urea, protein's major waste product¹⁶²; urea and other nitrogenous waste can accumulate and contribute to uremic symptoms. Steady-state BUN levels directly parallel levels of all nitrogen-containing waste products that contribute to uremic symptoms,¹⁶³ so following BUN is a helpful guide to the accumulation of these substances. Steady-state concentration of BUN is a function of dietary protein intake that contributes to urea production, of the comparatively small amount of urea degradation and extra-kidney clearance,¹⁶² and of how well the kidney excretes urea, the latter being directly related to GFR. It follows that reducing dietary protein is an effective strategy to reduce BUN and the associated blood levels of nitrogenous wastes in patients with declining GFR.

Metabolic acidosis, a comorbid condition that appears when eGFR is comparatively low (typically, <40% of normal),¹⁶⁴ increases protein catabolism¹⁶⁵ mediated through the adenosine triphosphate-dependent ubiquitin-proteasome system.¹⁶⁶ Correction of the metabolic

acidosis in patients with CKD with alkali decreases protein catabolism, improves protein balance with increased muscle mass, and decreases induction of the ubiquitin-proteasome system.^{167,168} Metabolic acidosis also decreases protein anabolism, manifest by decreased albumin synthesis¹⁶⁹ that is also improved with correction of the metabolic acidosis.¹⁷⁰ Because diets in developed societies are typically acidogenic³⁷ and can contribute to metabolic acidosis, particularly in individuals with low GFR,¹⁷¹ modifying diets can help prevent or treat metabolic acidosis in CKD and thereby ameliorate its untoward effects on protein metabolism. Options to accomplish this modification include reducing the amount of animal-based protein in the diet and/or adding base-producing fruits and vegetables, the latter of which effectively treats metabolic acidosis in CKD^{103,104,172} and reduces acid accumulation in patients with CKD, reduced eGFR, but no metabolic acidosis.¹⁷³

Guidelines for dietary protein in CKD have most often focused on the amount of dietary protein but less so on character of the ingested protein. The 2020 Kidney Disease: Outcomes Quality Initiative (KDOQI)¹⁷⁴ guideline 3.0.2 for CKD 3 to 5 patients, not on dialysis, suggests lowering protein intake to 0.6 to 0.8 g/kg body weight/day in adults with diabetes (OPINION) or without diabetes, under medical nutrition therapy (MNT); a low-protein diet providing 0.55 to 0.60 g dietary protein/kg bw/day; or a very-low-protein diet providing 0.28 to 0.43 g dietary protein/kg bw/day with additional keto acid/amino acid analogs to meet protein requirements (0.55–0.60 g/kg body weight/day) to reduce risk for ESKD/death (1A) and improve quality of life (2C level of strength, meaning that this is a suggestion and that if true effect may be substantially different from the estimate of the effect). Furthermore, despite the benefit of low-protein diets to reduce the rate of GFR decline shown in some studies, some patients with intakes below 0.8 g/kg/bw/day, even the described minimum level of 0.6 g/kg/bw/day, develop protein energy wasting¹⁵⁹ and have increased mortality.¹⁵⁶ Accordingly, these restrictive diets should be tailored to individual patients and under MNT guidance. The guidelines suggest that the recommended protein be of “high biologic value” but do not discuss if this includes plant-based protein. The ketogenic diet draws a majority (~70%) of its calories from fat, 15% from carbohydrates, and the remainder from protein. The long-term implications of these diets are not known and might include health risks such as increased net acid load, potential for dyslipidemia, and increased albuminuria.¹⁷⁵ Past guidelines suggested avoiding high protein intake (>1.3 g/kg/bw/day) in adults with CKD at risk of progression; nevertheless, the qualifications for these suggestions reflect the need for additional research to develop more concrete dietary protein recommendations for patients with very low (<30 mL/min/1.73 m²) GFR. In particular, further studies will elucidate the importance of the protein's character compared with its amount, particularly with respect to its acid- or base-producing capacity. In addition to the aforementioned potential advantages of plant-based compared with animal-based protein in CKD, carnitine and lecithin in animal-based protein are metabolized by gut microbes to trimethylamine, which in turn is metabolized by liver flavin monooxygenases to trimethylamine-N-oxide

(TMAO). High blood levels of TMAO strongly correlate with CVD and associated acute clinical events of CKD.¹⁷⁶ Together, these data support considering the character, in addition to the amount of dietary protein recommended for patients with CKD, particularly plant-based protein for those able to tolerate the increased potassium that accompanies plant-based diets.

- Sodium.** Adults in the United States consume an average of 3400 to 3600 mg (148–156 mmol) of sodium per day,^{177,178} an amount in great excess of dietary recommendations.¹⁵² Processed foods comprise the majority of sodium consumed in the United States. The food industry typically adds salt to increase shelf life and enhance palatability for consumers.¹⁷⁹ Foods eaten outside of homes typically have even higher sodium content.¹⁷⁹ Higher dietary sodium in CKD is associated with higher blood pressure, whereas its reduction is associated with lower blood pressure and albuminuria.^{180,181} As discussed, higher dietary sodium in patients with CKD is associated with more rapid CKD progression.¹¹² In addition, higher dietary sodium in patients with CKD is associated with attenuated kidney-protective effects of angiotensin-converting enzyme inhibitors¹⁸² and cardiovascular protective effects of angiotensin receptor blockers,¹⁸³ both mainstream therapies for CKD.¹⁷⁴ Together, these data support some level of dietary sodium restriction below current dietary intake in patients with CKD, but data to date remain insufficient to determine the optimal level of dietary sodium intake. The 2015 report of the American Dietary Guidelines Advisory Committee¹⁸⁴ recommends a daily sodium intake for adults of <2300 mg (100 mmol). The KDOQI recommendation (6.5.1)¹⁷⁴ for adult patients with CKD is a daily sodium intake of <2000 mg (90 mmol) unless otherwise contraindicated. Although this recommendation is of “2A” quality (i.e., moderate-quality evidence is a suggestion, but the true effect may be substantially different from the estimated effect), it seems reasonable until more definitive studies determine a more specific level of dietary sodium restriction.
- Potassium.** Patients with CKD appear to benefit from diets high in plant-based components including fruits and vegetables, as discussed. Nevertheless, the high potassium content of these diets historically has made practitioners reluctant to prescribe them because of the reduced potassium excretory capacity of patients with CKD who have reduced GFR.¹⁸⁵ Nevertheless, only limited data to support that dietary potassium restriction improves outcomes in CKD and most recommendations are opinion based. **Relatedly, some reports question the historic admonition to limit intake of plant-based foods in patients with CKD including fruits and vegetables.**^{185,186} Despite the understandable concern that higher dietary potassium could lead to hyperkalemia, there is little relation between dietary potassium and serum potassium in patients with CKD.¹⁸⁷ This lack of a close direct relationship between dietary and serum potassium is due in part to enhanced kidney excretion mediated by higher serum potassium, increased serum aldosterone and increased distal nephron urine flow in response, and enhanced Na^+/K^+ ATPase activity that helps maintain potassium balance.¹⁸⁸ Patients with CKD stage 4 who had fruits and vegetables added to their ad lib diets increased urine aldosterone excretion and exhibited urine hormonal changes consistent with lower urine 11β -hydroxysteroid dehydrogenase type 2 activity.¹⁷² The latter response to this increment in dietary potassium due to added fruits and vegetables indicates glucocorticoid access to the mineralocorticoid receptor, which along with aldosterone, yields added stimulation of distal nephron potassium excretion.¹⁸⁹ Importantly, potassium in plant-based food is predominantly in the form of nonchloride ions, a form that facilitates urine potassium excretion.¹⁹⁰ Healthy individuals given potassium with the nonchloride anion citrate had more pronounced RBC uptake and urine excretion of potassium after an acute oral load compared with an identical oral load of potassium chloride, while plasma potassium was similar.⁹¹ These data support the potential impact on potassium homeostasis of potassium intake with nonchloride anions like citrate that constitute the potassium content of plant-based diets. Additionally, patients with CKD demonstrate increased potassium secretion into the gastrointestinal system with increased fecal potassium losses.¹⁹² Together, these data support that patients with CKD have the capacity to increase potassium excretion in response to an increment in dietary potassium when given in the form of fruits and vegetables.
- A comprehensive analysis of 36,000 patients with CKD stages 3 to 4 reported a U-shaped mortality risk for both higher (>5 mEq/L) and lower (<3.5 mEq/L) serum potassium concentrations.¹⁹³ Additionally, a secondary analysis of 840 patients from the Modification of Diet in Renal Disease cohort found that higher urine potassium excretion, a surrogate marker for higher dietary potassium intake, was associated with a lower risk for all-cause mortality.¹⁹⁴ This higher potassium intake was unlikely due to pharmacologic potassium administration to these patients with CKD known to have reduced ability to excrete potassium; more likely, the higher potassium intake was due to higher intake of potassium-containing foods like plant-based food components that these patients tolerated without untoward effects. In support of this supposition, patients with CKD stages 3 to 4 tolerated diets high in plant-based food components given to improve calcium/phosphate metabolism^{195,196} and to treat metabolic acidosis^{103,104,172} without developing hyperkalemia. In addition, patients in these studies tolerated these plant-based diets despite their taking inhibitors of the renin-angiotensin-aldosterone system, which can limit urine potassium excretion, given for kidney protection. Limiting other drugs that can also reduce urine potassium excretion, such as nonsteroidal antiinflammatory drugs (NSAIDs) as was done in these studies,^{103,104,172,195,196} might help CKD patients eating a potassium-rich diet avoid hyperkalemia. In a cross-sectional evaluation of dietary recalls from patients on hemodialysis in the United States, the top sources of dietary potassium were meat products (beef, chicken, Mexican food, hamburgers), followed by legumes in fifth place.¹⁹⁷ While emphasis is often stressed upon patients to restrict high potassium plant foods, animal potassium sources often receive less attention from dietitians and practitioners.

- The 2015 report of the American Dietary Guidelines Advisory Committee¹⁸⁴ recommends a daily potassium intake for the general population of adults of 4700 mg (121 mmol). Nevertheless, average daily potassium intake for U.S. adults is 2155 mg (55 mmol), likely lower than recommended due to poor intake of fruits and vegetables.¹⁹⁷ The NKF-KDOQI initiative has recommended potassium intake identical to the general population in patients with CKD stages 1 and 2 (>4 g/day or >102 mmol/day) and reduced intake of potassium to 2 to 4 g/day (51–102 mmol/day) in patients with CKD 3 and 4.¹⁹⁸ Because of the described benefits of plant-based diets, recommendations to restrict potassium intake should ideally be tailored to allow patients with CKD and hypertension to eat the DASH diet, and to allow for ample intake of fruits and vegetables to promote lesser degrees of metabolic acidosis and improved management of bone and mineral metabolism relative to a diet higher in salt, animal protein, and low-fiber carbohydrates.^{103,104,172} Potassium-binding agents might allow for liberalization of dietary potassium intake in patients with CKD, particularly when given with renin-angiotensin-aldosterone system blockers,¹⁹⁹ and we appear to be moving toward a strategy to allow for such liberalization through use of these drugs.²⁰⁰
 - **Phosphorous.** Patients with CKD and reduced GFR are in positive phosphorus balance yet can maintain serum phosphate concentrations in the normal range until advanced CKD stages (i.e., stages 4–5) through phosphaturia induced by increases in fibroblast growth factor-23 and parathyroid hormone.²⁰¹ Phosphorus metabolism is linked to CVD, vascular calcifications, and morbidity in patients with CKD.²⁰² Nevertheless, dietary phosphate intake measured by 24-hour urine phosphate excretion was not associated with ESKD incidence or cardiovascular mortality as long as the resulting serum phosphate concentrations were within the normal range.²⁰³ Historically, practitioners have limited dietary phosphorus intake by limiting intake of phosphorus-containing foods, particularly processed foods that often have elemental phosphate-added preservatives.¹⁴⁵ The elemental phosphates added to processed foods are of particular concern because they are quickly and completely absorbed from the gastrointestinal tract when ingested.²⁰⁴ By contrast, phosphate in plant-based food is in the form of phytates (phytic acid) that have much lower gastrointestinal tract absorption than the elemental phosphate in food additives.^{198,205} This helps explain the lower levels of serum phosphate and FGF23 in patients with CKD eating a primarily plant-based diet in comparison with those eating a diet more typical of developed societies in which the protein source was primarily animal based.^{195,196} In addition, patients with CKD eating the plant-based diet needed fewer phosphate-binding medications, with a resulting lower pill burden in these patients with advanced CKD.¹⁹⁸
 - Together, these data support the recommendation for restricted phosphorous intake in patients with CKD, specifically to <800 mg/day¹⁴⁵ in comparison with the average of 1000 mg/day for the average U.S. adult.¹⁵² Importantly, this requires limited to no intake of processed foods that contain elemental phosphate additives.
- Relatedly, fresh and home-prepared foods yielded better serum phosphate control than processed foods without compromising nutritional status.²⁰⁶ Dietary phosphorus restriction can also be helped by restricting intake of animal-based protein and/or adding plant-based protein with its less absorbable phosphates as discussed^{198,206} and/or substituting plant-based for animal-based dietary protein.
- **Calcium.** Recommendations for daily dietary calcium intake are 1000 mg (23 mmol) for adults aged 19 to 50 years, 1200 mg (28 mmol) for women 51 to 70 years old, and 1200 mg for all adults older than 70 years for the general population.²⁰⁷ Published data are insufficiently robust to offer a specific recommendation for dietary calcium intake for patients with CKD beyond that for the general population.¹⁷⁴ Recognizing the potential benefits of a plant-based compared with an animal protein-based diet in CKD patients, such diets affect overall calcium metabolism. Individuals eating a primarily animal-based protein diet have higher urine calcium excretion than those eating a primarily plant-based diet.²⁰⁸ Earlier studies support that this higher urine calcium excretion appears due to bone buffering of acid produced from metabolism of the ingested animal-based protein with resorption of bone minerals and release of calcium into extracellular fluid.²⁰⁹ More recent studies support that a likely greater contributor to the increased urine calcium excretion observed with the diets high in animal-based protein is that such diets are typically low in oxalate that binds to calcium in the gastrointestinal tract, thereby preventing calcium absorption and promoting its fecal excretion.²¹⁰ Greater gastrointestinal calcium absorption promoted by these low oxalate diets leads to greater calcium transport into the extracellular fluid with greater increased urine calcium excretion.²¹¹ On the other hand, diets high in plant-based protein have high amounts of calcium-binding oxalate that yield decreased gastrointestinal calcium absorption with subsequent lower urine calcium excretion.²¹⁰ Providers should be mindful of selected medications commonly prescribed to patients with CKD that might alter the recommended dietary calcium intake. For example, patients with CKD treated with calcitriol or other active vitamin D analogs will normalize intestinal calcium absorption, whereas patients with diminished 1-hydroxylation of 25-hydroxy vitamin D may develop hypocalcemia despite adequate dietary intake, owing to diminished intestinal absorption. Loop diuretics augment urinary calcium excretion, whereas thiazide and thiazide-type diuretics reduce urinary calcium excretion and may lead to distinct effects on serum calcium concentrations.
 - **Dietary fiber.** Patients with advanced CKD appear to have a “leaky gut” with translocation of endotoxins and bacterial DNA with untoward downstream consequences including increased inflammation and increased CVD morbidity.²¹¹ Diets typical of developed societies drive a milieu of metabolic abnormalities including uremic toxin production, inflammation, and immunosuppression that ultimately promotes progressive kidney failure and cardiovascular disease.^{211,212} Some dietary components are metabolized in the colon by fermentation, with the two predominant types being saccharolytic (carbohydrate)

and proteolytic (protein). Saccharolytic fermentation is beneficial with generation of short-chain fatty acids such as butyrate, propionate, and acetate, whereas proteolytic fermentation produces uremic toxins such as p-cresyl sulfate and indoxyl sulfate.²¹³ Furthermore, diets with higher animal-based protein-to-fiber ratio are associated with higher serum levels of p-cresyl sulfate and indoxyl sulfate.²¹³ On the other hand, analysis of 14,543 NHANES participants demonstrated that increases in dietary fiber intake were associated with reductions in inflammation and mortality in patients with CKD.²¹⁴ Increased dietary fiber was associated with decreased mortality in CKD patients²¹⁵ and reduced serum concentrations of creatinine and urea nitrogen.²¹⁶ The many discussed benefits of increasing the dietary intake of plant-based dietary components in patients with CKD include increasing their fiber intake, which is generally lower than that of the general population.²¹⁶

- Mean daily fiber intake in the general U.S. population is 16 g, much lower than the recommended daily intake of 25 to 30 g.²¹⁷ On the other hand, mean daily fiber intake for patients with CKD was 12 g,²¹⁸ showing even lower intake than the general population, likely due to the historic limitation of plant-based foods in patients with CKD. The American Heart Association recommends daily fiber intake >25 g to help reduce CVD risk.²¹⁹ Because of the comparably high CVD risk in patients with CKD,⁵ it seems prudent to follow this recommendation for patients with CKD. Increased intake of plant-based foods is an effective strategy to increase dietary fiber, but this intervention should be limited to patients with CKD who can tolerate the associated increased potassium load without developing hyperkalemia.
- **Micronutrients.** Unfortunately, there have been few published studies regarding the micronutrient status of CKD patients and their need for trace elements and vitamins required for energy production, cell, and organ function, as well as for growth. Although some studies show that many patients on hemodialysis have dietary intakes of trace elements and vitamins that fall below recommended values,²²⁰ plasma levels of these substances in such patients are typically not low.²²¹ Other studies in a general population show that high intake of several micronutrients was associated with a reduced CKD risk.²²² Despite most guidelines that recommend multivitamin supplements for dialysis patients,¹⁴⁵ a comprehensive examination of the literature found insufficient evidence to support routine multivitamin use in patients receiving hemodialysis.²²³

Nevertheless, patients on peritoneal and hemodialysis are routinely exposed to dialysates without trace elements and because many patients with CKD are frequently prescribed diuretic agents, practitioners have understandable concerns that patients with CKD are at risk for being deficient in water-soluble vitamins. Some studies report low serum vitamin C concentrations in patients receiving peritoneal dialysis,²²⁴ and vitamin C supplements reduced dosage requirements for erythropoiesis-stimulating agents in other studies.²²⁵ Still others report that L-carnitine supplementation reduced resistance to erythropoietin-induced red blood cell production.²²⁶ Relatedly, half or more of nondialysis

patients with CKD had plasma studies consistent with iron deficiency.²²⁷ Accordingly, dietary guidelines for CKD patients often include supplements for the water-soluble vitamins including B₁, B₆, B₁₂, C, folate, and niacin.¹⁴⁵ Vitamin D is commonly prescribed to patients with CKD, and its appropriate use is discussed elsewhere. The paucity of published studies regarding the status of other fat-soluble vitamins in CKD patients does not allow an evidence-based recommendation for them.

Because of the disproportionately high contribution of coronary artery calcification (CAC) to the increased CVD risk of CKD²²⁸ and the contribution of vitamin K-dependent matrix Gla protein to the physiologic prevention of CAC,²²⁹ vitamin K status of CKD patients has attracted attention. Dietary intake of vitamin K was low in patients on hemodialysis in comparison with the general population^{230,231} and was inversely related to CVD-related mortality in nondialysis CKD patients.²³² Because many fruits and vegetables contain vitamin K,²³⁰ vitamin K supplementation has the potential to reduce vascular calcification in patients with CKD.

Regarding micronutrients in CKD, it seems that a reasonable approach would be to recommend the previously described components of a “healthy” diet as tolerated, emphasizing plant-based proteins, with a smaller proportion of animal-based protein to provide micronutrients that are more abundant in animal-based compared with plant-based food components such as riboflavin.

OVERALL DIET CONSIDERATIONS IN MANAGEMENT OF CHRONIC KIDNEY DISEASE

DIETARY APPROACHES TO PREVENT CONDITIONS THAT COMMONLY CAUSE CHRONIC KIDNEY DISEASE IN DEVELOPED SOCIETIES

As indicated, prevention of CKD importantly focuses on preventing the two major syndromes leading to CKD in developed societies, diabetes (predominantly T2D) and hypertension. [Tables 55.1 and 55.2](#) list strategic considerations in this approach for T2D and hypertension, respectively, and tactics to execute these strategies. Strategies common to both T2D and hypertension include diets that reduce net endogenous acid production, contain a high proportion of plant-based components, limit dietary NaCl, and reduce the risk for obesity. Institution of a predominately plant-based diet can achieve each of these strategies common to prevention of T2D and hypertension. This common thread of a dietary approach designed to prevent both T2D and hypertension supports a population approach emphasizing increased intake of plant-based foods. This approach holds promise to reduce the incidences of these two syndromes that contribute significantly to early death and increased disability in developed societies.¹ Such interventions, including this focus on increased intake of plant-based foods, are best applied at the level of overall populations if they are to reduce the incidences of T2D and hypertension and thereby CKD incidence. Doing so at the population level will likely require policy changes combined with public marketing. Practitioners can also consider these approaches for their individual patients deemed to be at

Table 55.1 Dietary Approaches to Reduce Incidence of Type 2 Diabetes

Strategy	Tactic
Emphasize carbohydrates that promote healthy glucose metabolism	- Promote foods with dietary fiber - Whole grains - Fruits and vegetables - Minimize foods with high glycemic index - Simple sugars - Fructose
Emphasize fats that promote healthy fat metabolism	Promote polyunsaturated fats, including omega 6's and omega 3's
Emphasize dietary protein that is less likely to increase insulin resistance	- Promote plant-based foods - Minimize animal-based foods, particularly red meat
Emphasize an overall diet that minimizes net endogenous acid production	- Promote base-producing fruits and vegetables - Minimize animal-based foods - Minimize NaCl ("salt")
Reduce obesity risk	- Promote a healthy caloric intake - Maximize foods with high-nutrition/low-calorie ratio like fruits and vegetables - Minimize processed foods - Promote omega-3 polyunsaturated fats

increased risk for T2D and hypertension as suggested by family history, ethnic history, and/or their socioeconomic circumstance.

DIETARY APPROACHES TO MINIMIZING THE RISK FOR INITIATION OF CHRONIC KIDNEY DISEASE IN SUSCEPTIBLE INDIVIDUALS/POPULATIONS

Practitioners can consider dietary approaches for individuals who presently have no CKD but are at increased risk for it in [Table 55.3](#). Their increased risk might be because they have diabetes (T2D or T1D) and/or hypertension, have other systemic conditions associated with CKD, or have a strong family history. Again, these strategies might be applied to overall populations considered to be at increased CKD risk (e.g., U.S. minoritized populations concentrated in under-resourced communities or indigenous Americans living on reservations).⁵ Primary care providers can also consider these interventions in patients with T2D and/or hypertension whom they manage.

DIETARY APPROACHES TO MINIMIZING THE RISK FOR PROGRESSION OF ESTABLISHED CHRONIC KIDNEY DISEASE TO END-STAGE KIDNEY DISEASE

[Table 55.4](#) depicts approaches that practitioners might consider reducing the risk for progression to ESKD for individuals with established CKD, particularly those with reduced eGFR at presentation. Successful prevention of, or delay of, ESKD adds to the quality of life of patients with CKD and reduces their management costs.⁵ Practitioners

Table 55.2 Dietary Approaches to Reduce Incidence of Hypertension

Strategy	Tactic
Emphasize a "healthy" profile of electrolyte intake	- Minimize NaCl ("salt") - Fresh rather than processed foods - Fresh fruits and vegetables - Promote potassium intake as tolerated - Fruits and vegetables - Promote magnesium intake as tolerated - Fruits and vegetables
Emphasize proteins that promote healthy blood pressure	- Promote plant-based foods - Minimize animal-based foods
Emphasize an overall diet that minimizes net endogenous acid production	- Promote base-producing fruits and vegetables - Minimize animal-based foods - Minimize NaCl ("salt")
Reduce obesity risk	- Promote a healthy caloric intake - Maximize foods with high nutrition/low-calorie ratio like fruits and vegetables - Minimize processed foods

Table 55.3 Dietary Approaches to Reduce the Risk for Initiation of Chronic Kidney Disease in Susceptible Individuals

Strategy	Tactic
Emphasize an overall diet that minimizes net endogenous acid production	- Promote base-producing fruits and vegetables - Minimize animal-based foods - Minimize NaCl ("salt")
Emphasize an overall diet with a high ratio of fiber-to-animal-based protein	- Promote plant-based foods - Minimize animal-based foods, particularly red meat
Emphasize fats that enhance kidney health	Promote polyunsaturated fats including omega 3's

currently underuse dietary strategies designed to prevent or slow CKD progression.¹⁰

DIETARY APPROACHES TO REDUCING MORTALITY IN ESTABLISHED CHRONIC KIDNEY DISEASE

Practitioners might consider the approaches listed in [Table 55.5](#) in an effort to reduce the mortality risk for individuals with established CKD, especially those with reduced eGFR because mortality in CKD is inversely associated with eGFR.¹³³ Practitioners managing patients with CKD often focus on CKD-related complications and less so on strategies to reduce mortality.¹³³ Currently, when the management focus includes mortality reduction, the focus is typically on pharmacologic strategies⁵ with less focus on the effectiveness of dietary strategies.¹⁰ As described for strategies to reduce CKD progression, dietary strategies designed to reduce mortality appear to be underused.

Table 55.4 Dietary Approaches to Reduce the Risk for Progression of Established Chronic Kidney Disease (CKD)

Strategy	Tactic
Emphasize an overall diet that minimizes net endogenous acid production	- Promote base-producing fruits and vegetables as tolerated - Minimize animal-based foods - Minimize NaCl ("salt") - Minimize NaCl ("salt") intake
Emphasize electrolyte intake that lowers risk for CKD progression	
Emphasize an overall diet that yields a gut microbiome which promotes kidney health	- Promote plant-based foods as tolerated - Minimize animal-based foods, particularly red meat

Table 55.5 Dietary Approaches to Reduce Mortality in Established Chronic Kidney Disease

Strategy	Tactic
Emphasize an overall diet with a high ratio of fiber-to-animal-based protein	- Promote plant-based foods - Minimize animal-based foods, particularly red meat - Maintain protein intake that avoids protein energy wasting
Minimize dietary NaCl ("salt")	- Promote fresh fruits and vegetables as tolerated - Minimize processed foods - Minimize NaCl ("salt") intake
Emphasize diets that improve blood pressure control	- Emphasize potassium intake as tolerated - Emphasize diets that reduce net endogenous acid production

Table 55.6 General Approach for a Dietary Recommendation for an Individual with Chronic Kidney Disease and Reduced Glomerular Filtration Rate

Dietary Consideration	Recommendation
Energy	30-35 kcal/kg body wt/day 5%-15% from carbohydrates - Avoid fructose
Protein	0.7-0.8 g/ideal body wt/day for nondialysis patients 1.2 g/ideal body wt/day for dialysis patients - Nonnitrogen ketoanalogues might be substituted for a portion of recommended dietary protein - Plant-based protein can be considered for patients able to tolerate the additional potassium load
Sodium	<2 g (90 mmol)/day unless otherwise contraindicated
Potassium	>4 g (102 mmol)/day for CKD stages 1 and 2 2-4 g (51-102 mmol)/day for CKD stages 3 and 4 - As instructed by provider for CKD stage 5
Phosphorous	<800 mg/day - Promote plant-based sources over animal-based sources
Calcium	- Minimize processed foods 1000 mg (23 mmol)/day for adults aged 19-50 years 1200 mg (28 mmol)/day for adult females aged 51-70 years 1200 mg (28 mmol)/day for adults >70 years
Fiber	≥25 g/day
Micronutrients	- Promote a primarily plant-based diet - Multivitamin supplement for dialysis patients

GENERAL DIETARY MANAGEMENT OF PATIENTS WITH CHRONIC KIDNEY DISEASE

Table 55.6 lists general management approaches to CKD patients, distinguished by important nutritional components.

DIETARY APPROACHES TO MANAGING COMMON CHRONIC KIDNEY DISEASE-RELATED COMPLICATIONS

Table 55.7 lists some effective dietary strategies for management of the indicated CKD-related complications. Practitioners might use these dietary strategies along with appropriate pharmacologic interventions.

WHOLE DIETS THAT MIGHT BE PRESCRIBED FOR PATIENTS WITH CHRONIC KIDNEY DISEASE

Patients with CKD have historically been counseled as to individual dietary components to maintain kidney and

overall health. While effective, this "component" approach can be daunting for patients to comprehend and practically challenging to implement. Many patients often find it easier to be asked to consider and follow whole diets with a pattern of established food components that are available in hardcopy handouts and/or online. Table 55.8 gives examples of whole diets that practitioners might consider offering to their patients.

ACID/BASE-PRODUCING CAPACITY OF SOME WHOLE DIETS AND COMMON FOODS

Ongoing research is showing the importance of the acid/base-producing capacity of food to health including kidney health. Given the discussed studies showing the apparent contribution of high acid-producing diets to the risks for diabetes, hypertension, incident CKD and its progression, and mortality, Fig. 55.2 compares the acid/base-producing capacity of various whole diets, compared with the diet

recommended by the U.S. Department of Agriculture. As shown, all but one of the common whole diets have greater acid-producing capacity than that recommended by the U.S. Department of Agriculture. Only the Mediterranean diet has less acid-producing capacity, and it is net-base producing.

CONSIDERATIONS FOR ENABLING A HOLISTIC DIET FOR PATIENTS WITH CKD

Many of the dietary recommendations for patients with CKD supported by published studies include increasing the plant-based component of their diets. Nevertheless, diets typical of developed societies are composed predominately of animal-based components,³⁷ as discussed earlier. Facilitating patients with CKD from eating these diets requires

Table 55.7 Dietary Interventions That Help in Management of Chronic Kidney Disease–Related Complications

Strategy	Tactic
Nephropathy progression	- Blood pressure control - Glycemic control in patients with diabetes - Reduce net endogenous acid production - Promote a “kidney-friendly” gut microbiome
Hypertension	- Minimize NaCl (“salt”) - Promote plant-based food as tolerated - Potassium as tolerated - Magnesium as tolerated
Disturbed calcium/phosphate metabolism	- Promote plant-based food as tolerated - Minimize processed foods
Metabolic acidosis	- Promote base-producing fruits and vegetables as tolerated - Minimize animal-based protein - Minimize NaCl (“salt”)
Increased protein catabolism	Treat metabolic acidosis with the described dietary approaches

practitioners, health system leaders, and even government leaders to address real and perceived barriers for patients to acquire and consume such diets, including expense (e.g., fresh fruits and vegetables are generally more expensive than packaged, processed foods) and accessibility (many underresourced communities have limited access to fresh fruits and vegetables).²³³ Once patients with CKD gain access to these foods, practitioners must simplify the approach to healthy food patterns and whole diet recommendations such that patients have a variety of food options to achieve food patterns that achieve intended goals.²³⁴ This simplified approach must include culturally sensitive instructions

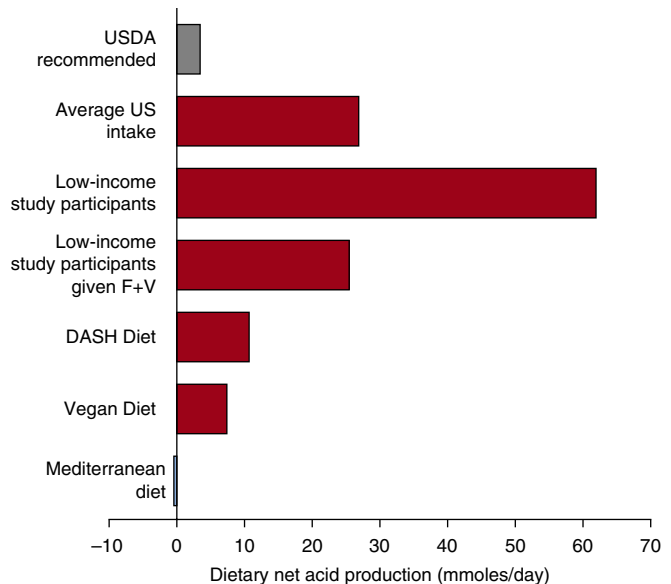


Fig. 55.2 Graphic presentation of the potential renal acid load, or net acid-producing capacity, of the indicated diets, calculated from their standard food components. The net acid-producing capacity of components of the U.S. Department of Agriculture (USDA) recommended diet is used as the standard to which to compare the net acid-producing capacity of the remaining indicated diets; it is represented by the *gray bar* at the top. Diets with net acid-producing capacity greater than the USDA diet are represented by *red bars*, and the one with less is represented by a *blue bar*. DASH, Dietary Approaches to Stop Hypertension; F + V, fruits and vegetables. (Data for diets of low-income study participants are from references 102, 103, and 171.)

Table 55.8 Common Components of Various Diets That Might Be Prescribed

	Mediterranean Diet	Mediterranean Kidney Diet	DASH Diet	Plant-Based Low-Protein (PLADO) Diet
Cereals/whole grains	Every meal	Every meal	7-8 servings/day	Every meal
Fruits + veggies	3-4 servings/day	K friendly F+V*	4-5 servings/day	3-5 servings/day
Olive oil	Every meal	Every meal	2-3 servings/day	2-3 servings/day
Fish/Seafood	5-6 times/week	5-6 times/week	0-2 servings/day	Minimal 14-32 g/day
Meat—white	0-2 times/week	0-2 times/week	0-2 servings/day	Minimal
Meat—Red, lean	<4 times/month	<4 times/month	0-2 servings/day	Minimal
Nuts/seeds	1-2 times/day	1-2 times/day*	1-2 servings/day	3-4 servings/day
Sweets	<3/week	<3/week	<5/week	3-5/week
Dairy	2 servings/day	2 servings/day	2-3 servings/day	2-3 servings/day

for food preparation.¹⁰³ In addition, gaining support of all members of a patient's household helps facilitate successful dietary interventions for patients with CKD whose recommended diet often contrasts with that of remaining household members.¹⁰³

DIET RECOMMENDATIONS

We suggest endorsing MNT-guided endorsements of kidney-friendly diets and particularly stress the benefits of Mediterranean-style dietary patterns. Dietary patterns may be easier to teach and follow than providing specific instructions to target individual foods and micronutrients. However, many gaps exist in knowledge regarding plant-based diets in the setting of CKD and further research is necessary to address these areas.

Studies show in cohorts of patients with CKD G3-5 that adherence to a Mediterranean diet,²³⁵ a DASH diet, or patterns of diet that are consistent with healthy plant-based diets is associated with lower CKD prevalence, lower risk of incident CKD, and slower progression to kidney failure.²³⁶ The 2020 KDOQI¹⁷⁴ guidelines also suggest recommending the Mediterranean diet in adults with CKD G 1 to 5 not on dialysis and in transplant patients to improve lipid control. The CORDIOPREV study (a prospective, randomized, controlled trial, including 1002 patients with coronary heart disease) evaluated the efficacy of 5 years of consuming a Mediterranean diet rich on kidney function. The intervention, compared with a low-fat diet rich in complex carbohydrates, was associated with slower eGFR decline in patients with coronary heart disease and T2D. Patients with a mild reduction in eGFR benefited more from the consumption of the Mediterranean diet in terms of preservation of kidney function with time.²³⁷ Additionally, studies from the laboratories of the authors have demonstrated that the Mediterranean diet pattern had the lowest acid-producing capacity (indeed it was net-base producing) among the diets studied (Fig. 55.2). Furthermore, plant-based diets play an important role in so-called “conservative” management of patients with CKD for whom the desire is to maintain a reasonable level of health and quality of life while avoiding dialysis.²³⁸

Below are the main characteristics of whole diets that might be considered for patients with CKD:

- Mediterranean diet: Dietary pattern rich in fruits, vegetables, grains, nuts, beans, olive oil, moderate to low amounts of fish, eggs, and poultry, and minimal amounts of red meat, sweets, and processed foods
- Dietary Approaches to Stop Hypertension (DASH) diet: Diet rich in foods with fiber, protein, fruits, vegetables, and low in salt
- Plant-Dominant Low-Protein (PLADO) diet: Daily protein intake of 0.6 to 0.8 g/kg/day with at least 50% plant-based sources to meet the targeted dietary protein which should preferably be whole, unrefined, and unprocessed foods.

FUTURE RESEARCH DIRECTIONS

Despite published data that support effective dietary approaches to the management of patients with CKD, many areas require further examination. For example, the

benefits to CKD patients of a plant-based diet on mortality, progression of underlying nephropathy, and some CKD-related comorbid conditions support such diets as first-line treatment of CKD patients with pharmacologic therapy as adjunctive.¹⁰ Future studies will determine, however, if this suggested approach will become standard of care for CKD management. In addition, future research will better determine in what circumstances these diets are most effective and how best to facilitate their implementation by patients with CKD. In the meantime, it is imperative that clinicians, especially those providing primary care, recognize the kidney protection benefits of the discussed dietary interventions for CKD patients.



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